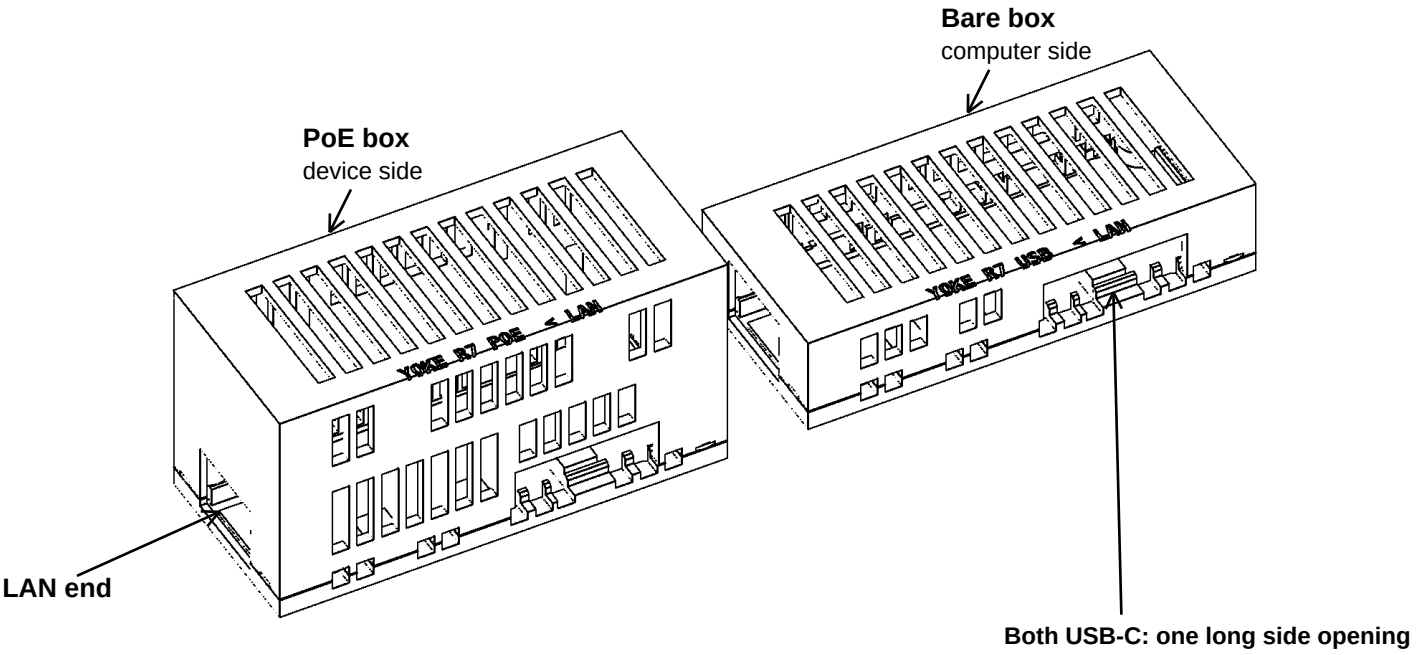
Know the parts and hardware

Two boxes for the ordered Radxa ZERO 3E 2GB WITH GPIO (Amazon B0CZD1C54N). Device side: ZERO 3E + Radxa PoE HAT (B0D1KF1L7G), powered over Ethernet. Computer side: bare ZERO 3E, powered by the computer through USB-C. Electronics have not arrived; verify them before fitting or powering.

PoE box 81.4 x 36.25 x 41.7 | Bare box 81.4 x 36.25 x 18.5



Opposite end closed. HDMI covered. microSD: lid off, lift board, change card.

1 Inventory and measure owner stock

Count four printed parts: two identical bases (YOKE R7 BASE under the floor), the PoE lid (YOKE R7 POE) and the bare lid (YOKE R7 USB). Count eight M3 x 8 socket-head screws and eight RX-M3x5.7 inserts. Measure screw length, head envelope and insert stock before heating.

Stop for wrong stock, heads over 6.0 x 3.0, damaged threads or inaccessible tools. M3 x 16 and M3 x 20 are too long here.

Nominal screws: M3 x 8 socket head (8), measured under head; head at most diameter 6 x height 3. Inserts: RX-M3x5.7 (8), diameter 4.6 x length 5.7, printed pilots diameter 4.0. Four of each per box. No clips, standoffs or nylon screws are used.

PoE box only: Radxa's 4 insulated copper pillars and 8 CM2.5x4 screws. They carry PoE voltage (up to 57 V): keep them installed, tightened and insulated. No case fastener replaces or touches them.

Tools: caliper/depth gauge, 2.5 mm hex driver reaching 20 mm, Radxa screw driver, controlled insert heater/tip, nonflammable support, actual cables, temperature probe and CPU monitoring.

Source allowance: official RS109_RA023 CAD, 18 mm board gap, 1.0 roof clearance. Radxa's 14.5 mm pillar listing is unresolved; if the measured gap differs, only the PoE lid is regenerated and reprinted.

Fit empty; heat-set eight inserts

2 Test each empty lid on a base first

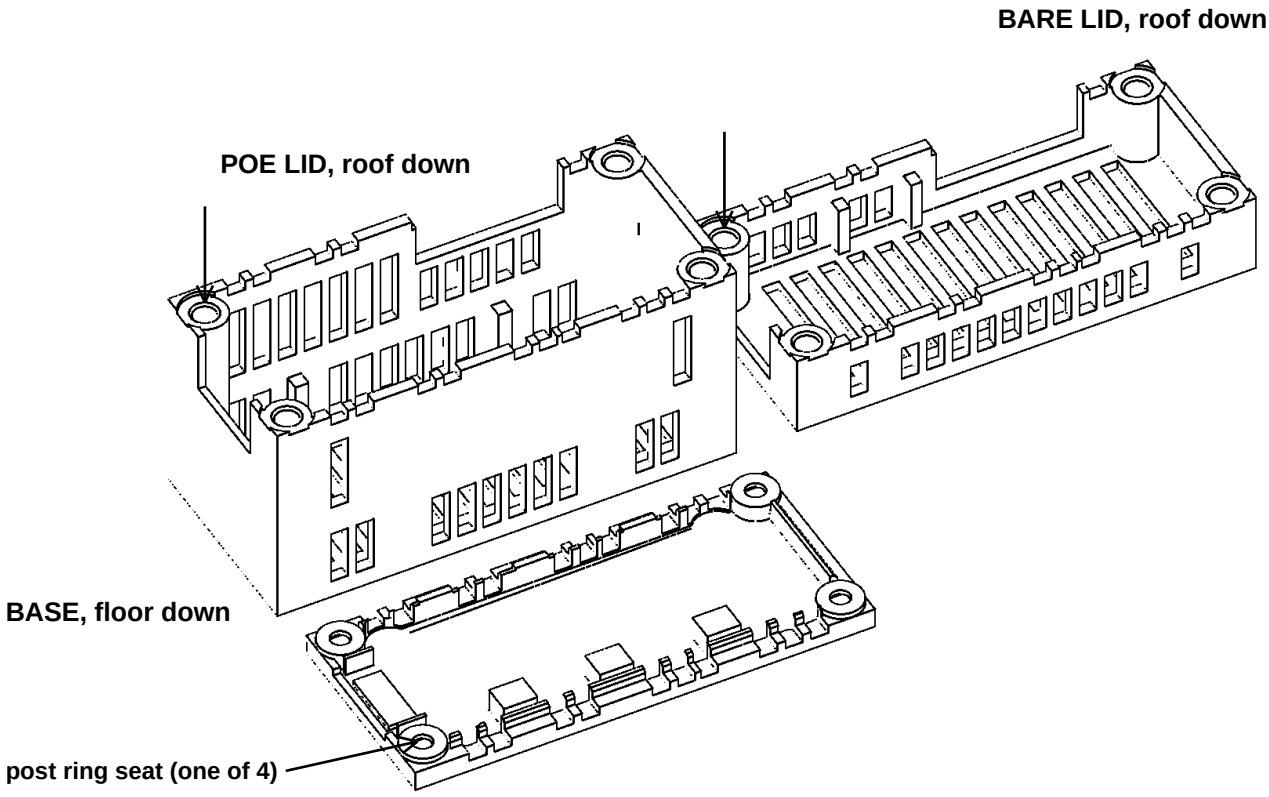
With no electronics, inserts or screws, match the LAN ends and lower each lid vertically onto a base. It must slide over the inner lip by hand and seat on the four corner post rings, leaving an even 0.3 shadow gap along the seam; then lift off freely. Check the actual driver in all eight bottom wells and the heater in all eight lid post pilots.

Stop for binding, rocking or skew. Screws must never pull a poor fit closed.

3 Heat only the empty lids, roof down

Support each EMPTY lid roof-down on a flat nonflammable surface. Heat-set one insert into the pilot at the bottom face of each of the four corner posts, straight down, finishing 0.2 below that face. Follow the insert maker's controlled-heater method. Cool fully; inspect squareness, threads and distortion.

Stop for a tilted insert, an insert proud of the post face, crack, bulged post or blocked thread. Never heat plastic with electronics installed.



Vertical arrows: one of the four corner post pilots in each lid; set an insert into each. In the closed box each insert spans Z = 4.8 to 10.5; the lid post face at Z = 4.6 seats on the base post ring. Let parts cool before threading screws or adding electronics.

Nesting joint: lid walls slide over a 1.2 inner lip with 0.4 clearance per side, 1.5 engagement and a 0.5 lead-in. Each solid base post ring (diameter 7.8) rises into a 8.6 seat recess in the lid. Lower and lift vertically; there is no hook, clip or sideways motion.

The lid rests on the post rings, so a 0.3 shadow gap stays at the outer seam by design. After cooling, repeat the empty fit. Never drive a screw to close the gap, flatten a tilted insert or reshape a post.

Nominal first article. Physical fit, durability and powered cooling: NOT RUN.

Dimensions in mm, Z from the base underside. Actual exported STL sections; nominal hardware envelopes, no threads.

PoE box: Radxa stack first, then seat

4 Assemble the Radxa stack outside the case

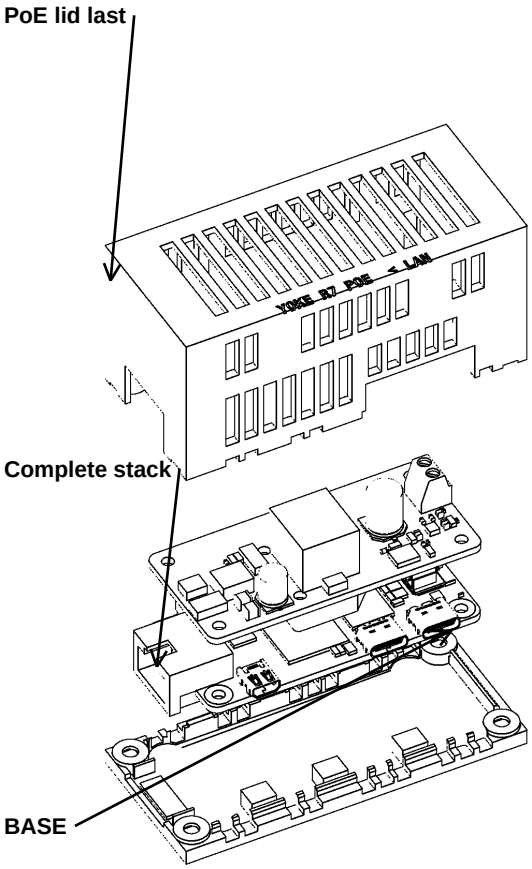
PoE box. Disconnect all power and cables. Outside the case, fit the microSD card, then follow Radxa's instructions: four lower CM2.5 screws into the four insulated copper pillars, HAT pressed fully onto GPIO, four upper CM2.5 screws. Tighten as Radxa requires. These conductive joints carry PoE voltage and stay installed.

Stop if the delivered SBC/HAT, pillars, insulation or electrical joints differ from Radxa's assembly or the reference.

5 Lower the stack into a base

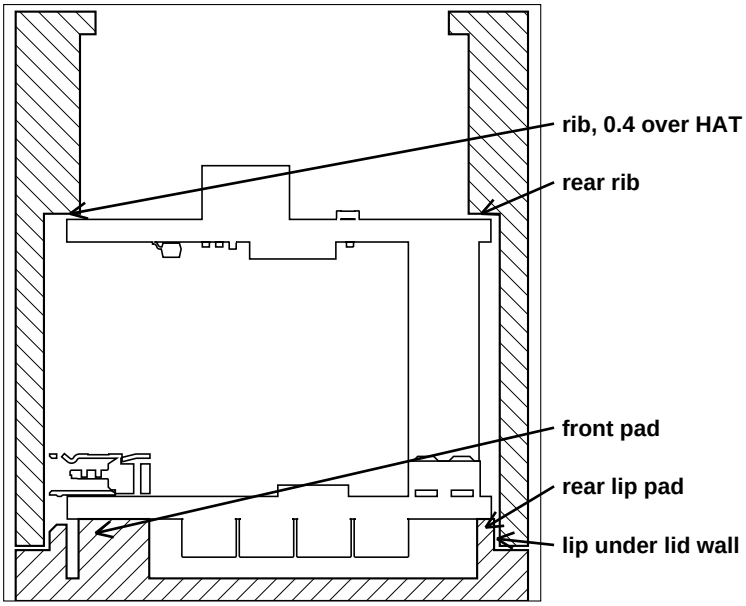
Lower the complete stack level into one base: RJ45 between the two LAN-end posts and into the low cradle, USB-C toward the long open side. The SBC underside rests on three front pads and three raised rear-lip pads; no factory screw or pillar touches plastic.

Stop for rocking, a bowed PCB, a component bearing on the case or plastic touching Radxa hardware.



B-B / PoE retention / X = 14

Y horizontal (USB-C side left), Z up



Actual cut through base, PoE lid, SBC and HAT meshes; board meshes are drawn as cut lines. Ribs bottom at Z = 27.4, 0.4 above the HAT top edge; the SBC underside rests on pads at Z = 5.8. Ribs limit lift only; they do not clamp the boards or load the pillars.

Fan is a reference feature; the ordered HAT may be fanless. Radxa assembly: docs.radxa.com/en/zero/zero3/accessories/3e-poe-hat

Nominal first article. Physical fit, durability and powered cooling: NOT RUN.

Dimensions in mm, Z from the base underside. Actual exported STL sections; nominal hardware envelopes, no threads.

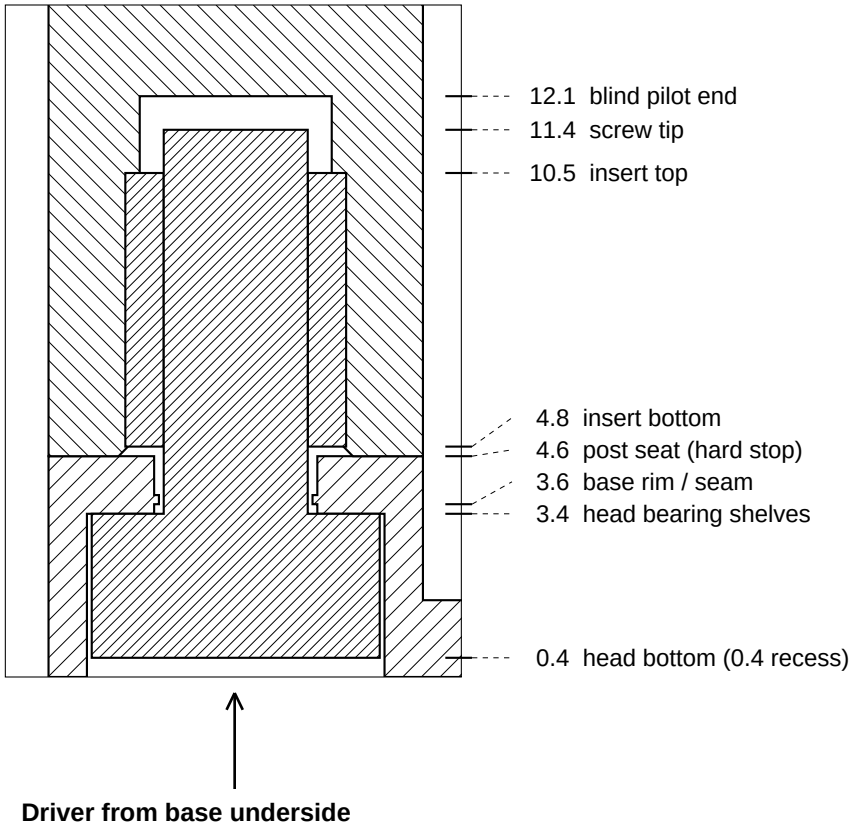
Close freely; screw from below

6 Close, hold together, invert, fasten

Match LAN ends and lower the PoE lid straight down over the stack. It must seat on the four post rings with the even shadow gap; its four inner ribs stop 0.4 above the HAT edges. Hold both halves together, invert, start all four M3 x 8 screws from the bottom wells, then lightly snug opposite corners. Return floor-down.

Stop for binding, roof or rib contact, cross-threading, spinning inserts or proud heads, or if a head shelf or post ring turns, cracks or separates while snugging. Never use screw force to close a gap.

A-A / case screw / Y = 1.2 / X horizontal, Z up



Four M3 x 8 per box. Head bears at Z = 3.4 on two bridged shelves, 0.4 above the bottom face. Full 5.7 insert overlap; the tip ends 0.9 beyond the insert with 0.7 blind-pilot reserve.

Load path: head -> bridged shelf fused into the solid base post ring (4 full layers above two bridge layers) -> post seat at Z = 4.6, about 29 mm2 -> lid post -> insert -> screw. The outer seam keeps its shadow gap; neither PCB is in this clamp stack.

Hand-tight only: no torque is validated. Start all four screws before snugging opposite corners. Stop for a proud head, spinning insert, a shelf or post ring that turns or separates, a bowed case or changed PCB support.

Section is an actual cut through the released base, PoE lid, screw and insert STL meshes. Base and lid have opposing hatches; metal has close hatching. The bare box uses the identical joint.

Hold both halves together and turn the case upside down so the driver is above the four bottom wells. The bottom face is flat; screw heads sit below it.

Computer-side box: bare board

7 Seat the bare board in the second base

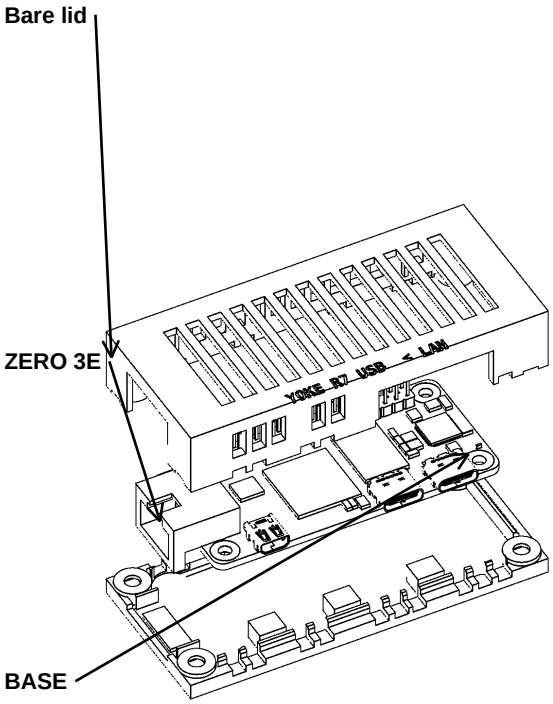
Computer-side box, no HAT. Fit microSD outside the case. Lower the bare ZERO 3E level into the second base: RJ45 between the LAN-end posts, USB-C toward the long open side, PCB on the six edge pads. Leave all four mounting holes empty.

Stop for rocking, a bowed PCB or anything entering a mounting hole; those holes connect to the RJ45 PoE centre taps.

8 Close and fasten like page 4

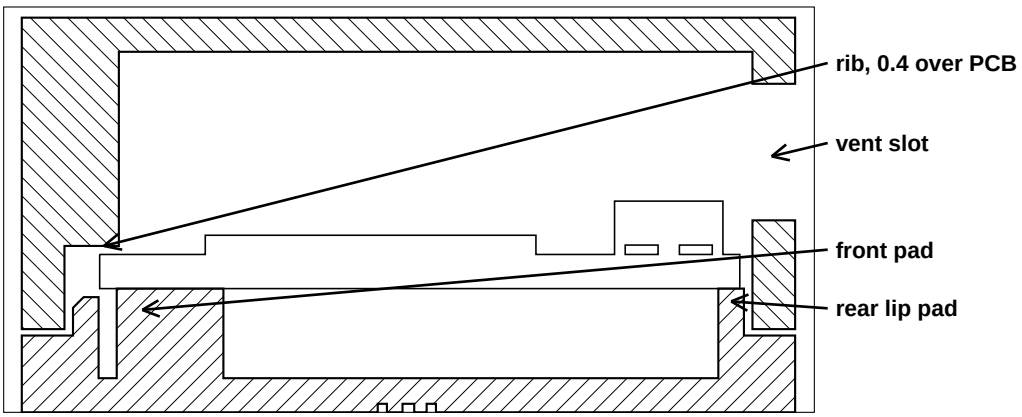
Match LAN ends and lower the bare lid straight down. Its four ribs stop 0.4 above the PCB edges. Hold together, invert, fit four M3 x 8 screws from below, lightly snug opposite corners.

Stop for binding, rib or roof contact with the board, cross-threading, proud heads, or a head shelf or post ring that turns, cracks or separates while snugging.



C-C / bare retention / X = 30.5

Y horizontal (USB-C side left), Z up



Actual cut through base, bare lid and official SBC mesh. Ribs at X = 16.2-18.2, 29-31, 5.95-6.85, 58.15-59.05 all stop at Z = 7.8 over component-free bare PCB edges, clear of the mounting-hole pads. The base is identical to the PoE box base.

Power and data: the computer's USB-C cable powers this box. Mounting holes stay empty; no peg, screw or metal reaches them.

Nominal first article. Physical fit, durability and powered cooling: NOT RUN.

Dimensions in mm, Z from the base underside. Actual exported STL sections; nominal hardware envelopes, no threads.

Service and first-article checks

9 Power off; reverse the installation order

Disconnect power and all cables. Hold the case inverted, remove the four bottom M3 x 8 screws, return floor-down and lift the lid straight up. Lift the complete board (or PoE stack) level out of the base; change microSD outside the case. Reassemble in page 3-5 order.

Do not pry on a PCB or connector. Do not loosen the Radxa conductive joints merely to open the case.

All physical checks below: NOT RUN. Complete after parts arrive.

A Inspect unpowered fit and cable access

Unpowered: record stock measurements, then accept only if the seam gap is 0.15-0.45 on every side, the measured HAT gap is 17.7-18.3 (otherwise set hat_gap and reprint only the PoE lid), heads sit at or below the floor and ribs clear the boards. Seat the actual Ethernet and both USB-C plugs; record play/rattle in use positions.

Stop for forced plugs, partly seated USB-C, cable strain, loose inserts, PCB bowing or component contact.

B Prove the service sequence ten times

Unpowered and unplugged, perform ten complete service cycles on each box, including board removal and microSD exchange. Each lid must seat freely before screws. Record seam gap, thread condition, insert movement and PCB play.

Stop at the first new bind, damaged thread, loose insert or cracked plastic; return that interface to design.

C Compare open and closed temperatures

After unpowered checks pass, run each box on a hard flat desk with its real power source (PoE switch or computer USB-C) and workload. Every 5 minutes record ambient, CPU and case temperatures, throttling and resets. Compare lid-open and lid-closed runs at least 30 minutes each, until three readings change under 1 C.

Stop for throttling, resets, odor, softening, any maker's limit or case surface above 45 C (provisional PLA limit, not a rating). No plateau leaves the test incomplete.

Air paths: low side inlets that stay open on a desk, side rows between and above the boards, and roof slots. Vent area and roof clearance do not establish cooling performance or a continuous power rating. Record fan presence on the PoE HAT.

Record before deciding to use these enclosures

Date / material / measured stock: _____

Delivered board + HAT / fan / board gap / fit + cables: _____

Cycles / seam gap / play / inserts / failures: _____

Ambient / workload / open vs closed temperatures / time: _____

Outcome / incomplete checks / next design change: _____